Forecasting Daily Product Views of an Online Clothing Store from Clickstream Data

Machine Learning Course Project – *e-shop clothing 2008* dataset

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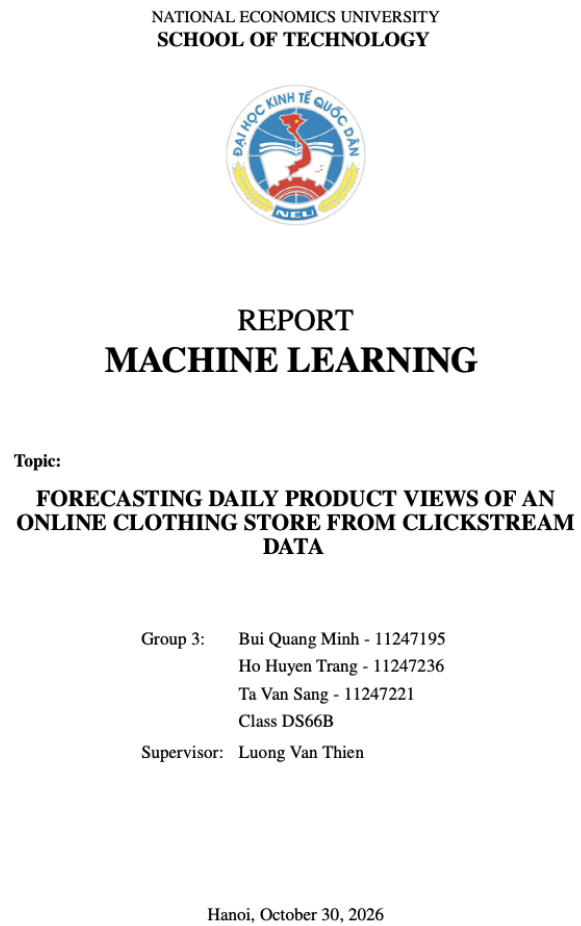
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Outline

1. Problem Statement
2. Existing Methodologies
3. Proposed Method
4. Binary Classification View
5. Experiments
6. Conclusion



NATIONAL ECONOMICS UNIVERSITY
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REPORT
MACHINE LEARNING

Topic:

FORECASTING DAILY PRODUCT VIEWS OF AN ONLINE CLOTHING STORE FROM CLICKSTREAM DATA

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Hanoi, October 30, 2026

Figure: Project Report

Problem Statement

Challenge: Every evening an online clothing store decides which products to feature and promote tomorrow, without knowing tomorrow's interest in each product

Key Issues:

- Public notebooks on this dataset mostly classify `price2`: high accuracy, but no business value
- The log has 165,474 clicks on 217 products and **no purchases**: daily views are the demand signal we can measure
- Leakage risk: any input that uses day d makes the forecast for day d look better than it is
- Small count target with many zeros (median 4 views, 10.2% zeros)
- Short history ($\approx$ 4.5 months) and a 12-day test window

Solution: Forecast tomorrow's views of every product (regression) and compare the course algorithms with strong naive baselines under a strict time-based protocol

Problem Formulation

Target for product p ($P = 217$) on day d:

$$y_{p,d} = \text{number of clicks on the page of product } p \text{ on day } d \tag{1}$$

Task: learn one global model f for all products

$$\hat{y}_{p,d} = f(\mathrm{x}_{p,d}), \qquad \mathrm{x}_{p,d} \text{ computed only from } \mathcal{F}_{d-1} \tag{2}$$

$\mathcal{F}_{d-1}$: everything known at the end of day $d-1$ (past views, product attributes, calendar)

Table: Inputs and outputs of the forecasting problem

Component	**Description**
Unit of observation	One (product, day) pair
Input	Lags and rolling means of past views, day of week, product attributes
Output	Number of views of the product on the next day
Problem type	Supervised regression, one-step-ahead forecasting
Main metric	MAE (plus RMSE, MAPE on $y > 0$, R^2)
Key constraint	Time-ordered split; features use only the past

Related Work Overview

Retail Demand Forecasting:

- Naive, seasonal naive, moving average [2]
- Per-series models: exponential smoothing, ARIMA
- M5 competition: global ML models with lag and rolling features [3]
- Limitation: simple benchmarks are hard to beat; leakage [4], random CV [5] and MAPE with zeros [6] are common pitfalls

Public Work on This Dataset:

- Dataset: Lapczynski & Bialowas [1], UCI repository
- Common task: classify `price2` (price above the category average) for each click
- Inputs: product attributes; very high reported accuracy
- Limitation: target leakage, memorisation of 217 products, no business use

Our Contribution: reformulate the task as daily view forecasting and evaluate it against the strongest naive baseline, with a leakage stress test

Re-examining the Common Approach: price2

Observation: each product keeps one price, colour, photo, page and price2 for five months

Table: Decision tree accuracy when predicting price2

Exp.	Setting	Accuracy
1	With price, random split of clicks	100.00%
2	Without price, random split of clicks	92.91%
3	Without price, tested on unseen products (5 groups)	44.1%
–	Always predict the majority class	**51.2%**

Key Findings:

- Exp. 1: price2 is computed from price and category ⇒ target leakage
- Exp. 2: the same products are in train and test ⇒ memorisation
- Exp. 3: on unseen products (44.1%, range 27.1–57.8%) it is **below** the majority rule
- The shop already knows its own prices ⇒ no business value

Proposed Method

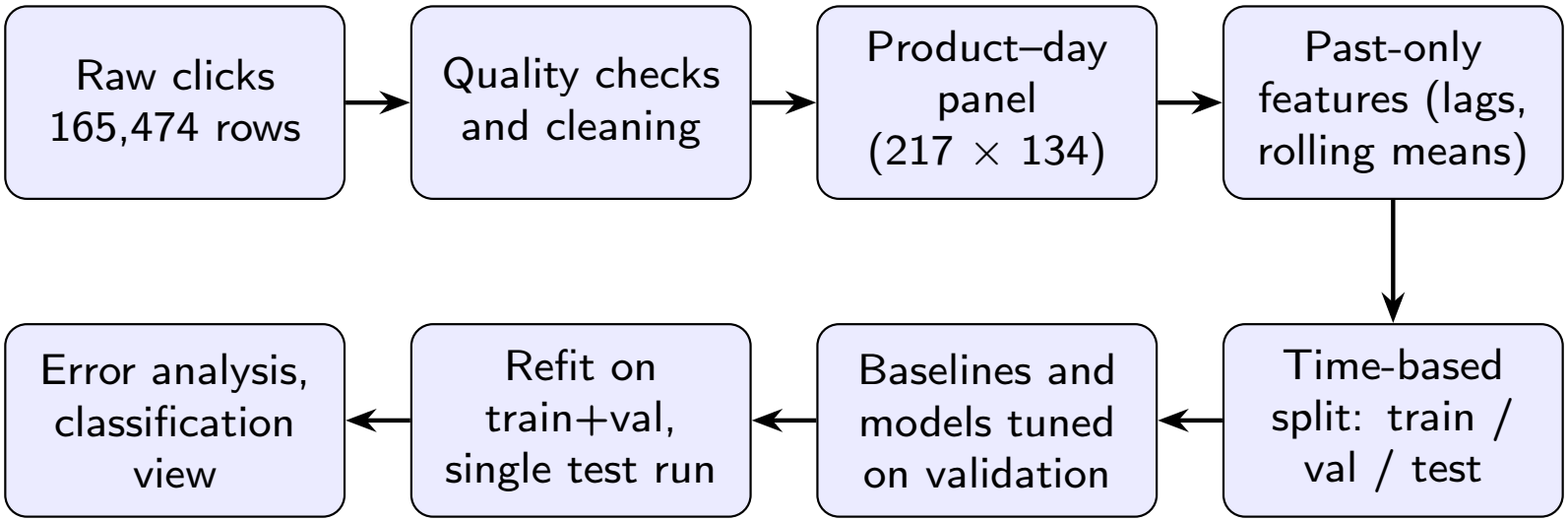


Figure: Overall pipeline: data preparation (top), modelling and evaluation (bottom)

Design rules:

- Features use only information available before the forecast day
- Split by time, never at random; scalers fitted on Train only
- All choices are made on validation; the test set is used **once**

Pipeline Components

Key Components:

1. **Data Cleaning**: remove the truncated last day; fix the category of product A18
2. **Product–Day Panel**:
   - One row per product and day; days without clicks get `views` $= 0$ (true zeros)
   - Exploratory analysis on the training period only
3. **Past-Only Features**:
   - Lags and rolling means computed within each product, always after `shift(1)`
   - Day of week and product attributes, one-hot encoded (38 columns)
4. **Models**: 4 baselines, Linear Regression, Ridge, KNN, Decision Tree, Random Forest
5. **Evaluation**: time-based split, tuning on validation, one final test run, reliability and error analysis

Data Quality Checks and Cleaning

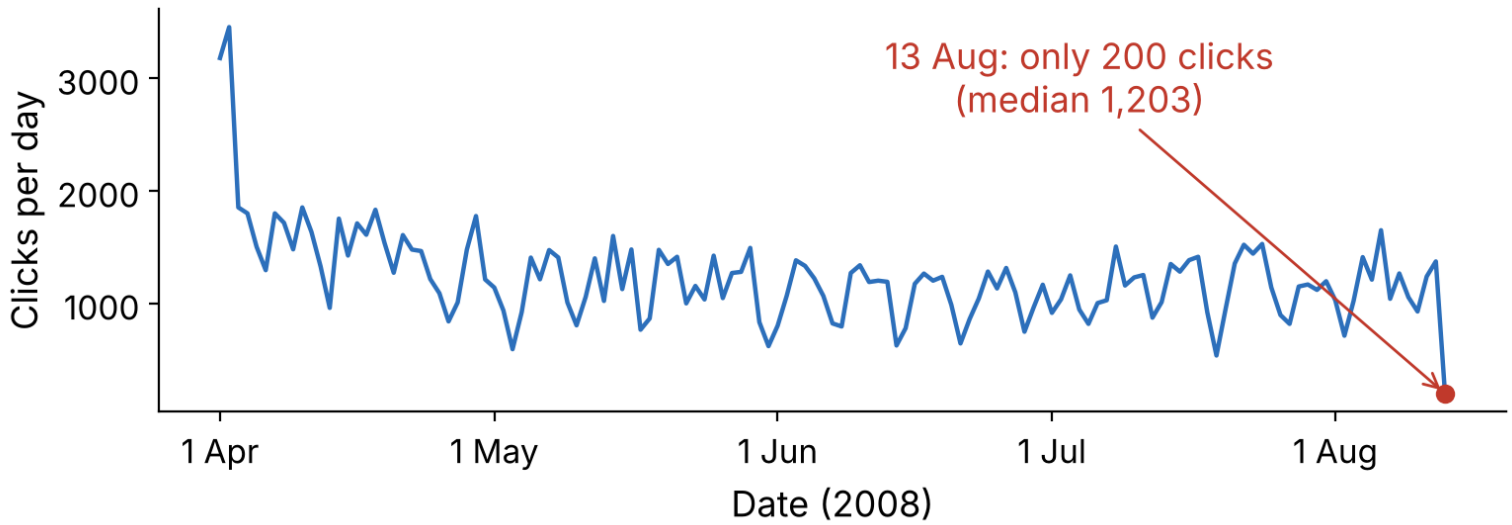


Figure: Total clicks per day, 1 April – 13 August 2008

- **Truncated last day**: 13 August has only 200 clicks ($\approx$ 17% of the median) $\Rightarrow$ removed, leaving 165,274 rows (134 days)
- **Inconsistent label**: A18 is in category 1 935 times and in category 2 once; all "A" codes are category 1 $\Rightarrow$ set to 1
- No missing values, no duplicated rows, no missing calendar days

Exploratory Analysis I: Target and Weekly Cycle

Training period only (up to 15 July): the validation and test periods stay unseen

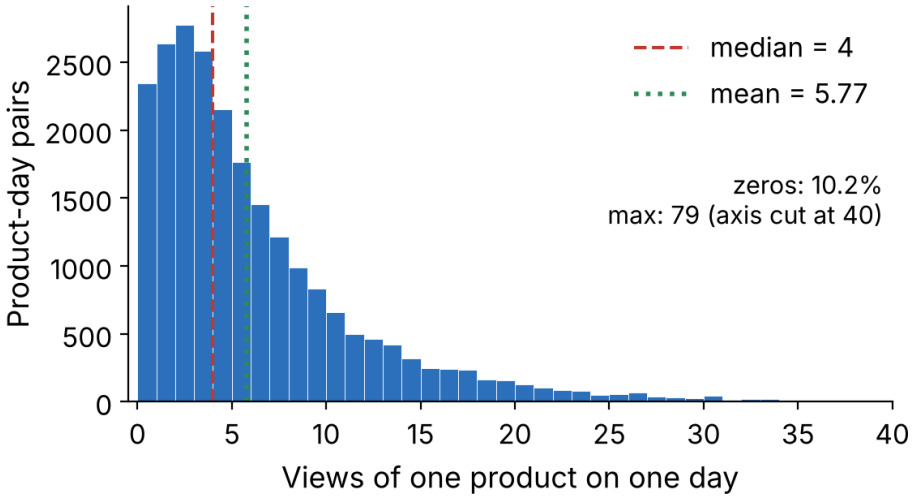


Figure: Distribution of `views`

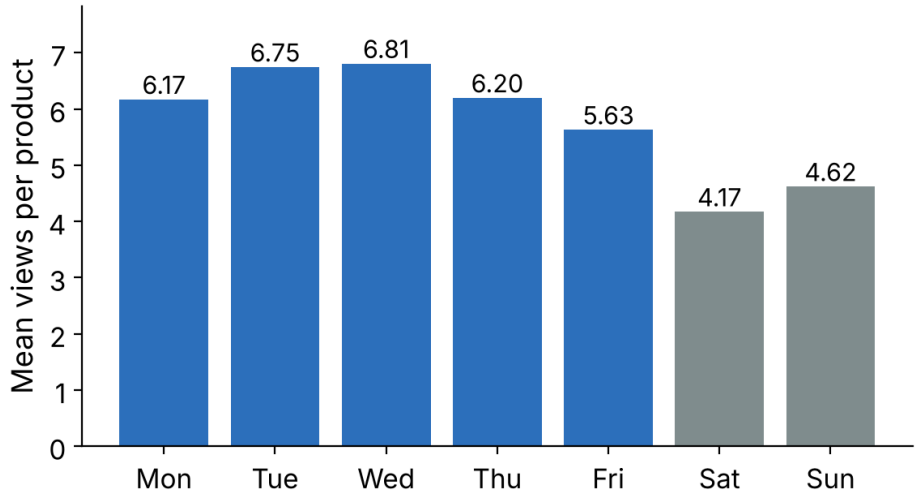


Figure: Mean views by weekday

- Right-skewed count: median 4, mean 5.77, max 79
- 10.2% zeros $\Rightarrow$ MAE as the main metric, MAPE only on $y > 0$

- Peak on Tue/Wed (6.75, 6.81), low on Sat/Sun (4.17, 4.62)
- $\Rightarrow$ features `day_of_week` and `lag_7`

Exploratory Analysis II: Category, Page and Confounding

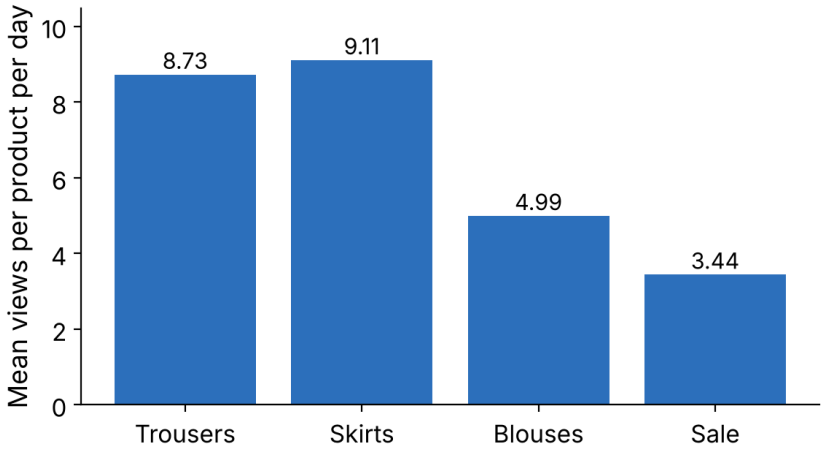


Figure: Mean daily views by category

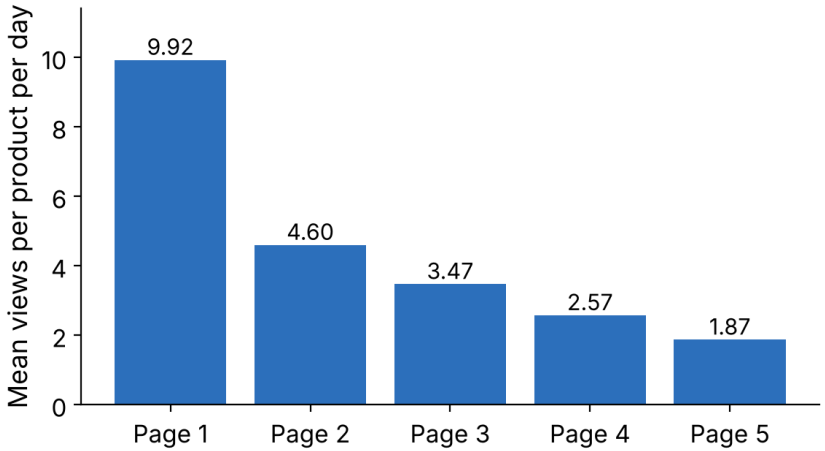


Figure: Mean daily views by page

Table: Products per category and page

Category	P1	P2	P3	P4	P5
Blouses	18	18	18	5	0
Sale	18	18	17	19	10
Skirts	17	16	0	0	0
Trousers	18	18	7	0	0

Insight: all skirts are on pages 1–2, sale items on all five pages ⇒ category is confounded with display position. Useful for prediction, but no causal claim

Feature Engineering and Leakage Control

History features for product p on day d (computed within each product):

$$\texttt{lag_1} = y_{p,d-1}, \quad \texttt{lag_7} = y_{p,d-7}, \quad \texttt{avg_k_days} = \frac{1}{k}\sum_{l=1}^{k} y_{p,d-l} \tag{3}$$

Implementation ($k \in \{7, 14\}$): always shift by one day *before* averaging

```
grp = panel.groupby('product')['views']
def avg_last_7_days(x):
    return x.shift(1).rolling(7).mean()
```

Manual check (product A1, 20 April):

- Views 13–19 April: $(12, 27, 19, 23, 16, 17, 11)$, mean $= 17.857$
- Notebook: avg_7_days $= 17.857$, lag_1 $= 11 \Rightarrow$ no leakage

Warm-up: drop the first 14 days $\Rightarrow$ 26,040 rows (217×120 days)

Time-Based Split and Models

Table: Time-based split of the modelling table

Set	From	To	Rows
Train	15 April 2008	15 July 2008	19,964
Validation	16 July 2008	31 July 2008	3,472
Test	1 August 2008	12 August 2008	2,604

Table: Models, inputs and hyperparameter grids (`random_state=42`)

Model	Inputs	Grid (chosen on validation)
0a–0d. Baselines	`lag_1`, `lag_7`, 7-day and 14-day averages	none
1a. Linear Regression	`lag_1` only	none
1b. Linear Regression	38 columns, standardised	none
2. Ridge	38 columns, standardised	$\alpha \in \{0.1, 1, 10, 100, 1000, 10000\}$
3. KNN	3 feature sets, standardised	$k \in \{10, 30, 50, 100, 200\}$
4. Decision Tree	38 columns	depth $\in \{3, 5, 6, 7, 10, 12, 15\}$, leaf ≥ 20
5. Random Forest	38 columns, 100 trees	depth $\in \{6, 8, 10, 12\}$, leaf ≥ 20

One-hot encoding (`drop_first=True`) learns nothing from data; the `StandardScaler` is fitted on Train only

Models: Key Equations

Linear and Ridge Regression:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^{m} \beta_j^2, \qquad \hat{y}_i = \beta_0 + \sum_{j=1}^{m} \beta_j x_{ij} \tag{4}$$

K-Nearest Neighbours (Euclidean distance on standardised features):

$$\hat{y}(\mathrm{x}) = \frac{1}{k} \sum_{i \in N_k(\mathrm{x})} y_i \tag{5}$$

Decision Tree (CART) and **Random Forest** ($B = 100$ trees):

$$\min_{j,t} \Big[\sum_{i \in L} (y_i - \bar{y}_L)^2 + \sum_{i \in R} (y_i - \bar{y}_R)^2 \Big], \qquad \hat{y}(\mathrm{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathrm{x}) \tag{6}$$

Negative predictions are clipped to 0 (views cannot be negative)

Evaluation Metrics

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{7}$$

$$\text{MAPE} = \frac{100\%}{|I|}\sum_{i\in I}\frac{|y_i - \hat{y}_i|}{y_i}, \qquad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{8}$$

where $I = \{i : y_i > 0\}$ (MAPE is undefined for zero actuals)

- **MAE** is the main metric: it is in views ("off by about two views")
- MAPE is inflated by small actuals (1 vs 2 views = 100% error)

Worked example used to test the `evaluate` function:

- Actual $(10, 5, 0)$, predicted $(12, 4, 1) \Rightarrow$ MAE $= 1.333$, RMSE $= 1.414$, MAPE $= 20\%$, $R^2 = 0.88$

Model Selection and Evaluation Protocol

Algorithm 1 Model selection and evaluation protocol

1: **for** each model M and each value h of its grid **do**
2: Fit M_h on Train; compute $\mathrm{MAE}_{\mathrm{val}}(M_h)$
3: **end for**
4: For each model keep $h^* = \arg\min_h \mathrm{MAE}_{\mathrm{val}}(M_h)$
5: Select the model and the baseline with the lowest validation MAE
6: Refit every model with h^* on Train $\cup$ Validation (scalers refitted)
7: Evaluate every model **once** on Test
8: Report the improvement of the selected model over the strongest baseline

Why refit? Train $\cup$ Validation (23,436 rows) adds the 16 days closest to the test period

Binary Classification View

Question: will the product be of *high interest* tomorrow?

- Label $= 1$ if `views` $\geq \tau = 7$ (75th percentile of the training target)
- Positive share: 29.4% (train), 28.7% (validation), 28.8% (test)
- Always "no": 71.3% Accuracy but F1 $= 0 \Rightarrow$ **F1** is the selection metric

Classifier Selection (lets us use the course classifiers properly):

$$\text{Logistic Regression:} \quad P(y = 1 \mid \mathrm{x}) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \sum_j \beta_j x_j \tag{9}$$

$$\text{Gaussian Naive Bayes:} \quad P(y = c \mid \mathrm{x}) \propto P(y = c) \prod_j P(x_j \mid y = c) \tag{10}$$

- Also KNN, Decision Tree and Random Forest classifiers (hyperparameters from Part 6)
- References: the rule "14-day average $\geq \tau$" and the RF regressor + threshold

Experimental Setup

Dataset: e-shop clothing 2008 (UCI)

- Clickstream of an online store for pregnant women, 1 April – 13 August 2008
- One row = one click; no purchases are recorded

Table: Size of the data at each stage of the pipeline

Stage	Attribute	Value
Raw log	Rows × columns	165,474 × 14
	Days / products / sessions	135 / 217 / 24,026
Panel	Product–day pairs	29,078
	Pairs with ≥ 1 click	26,127
Modelling	Rows after warm-up / features	26,040 / 38
	Train / validation / test rows	19,964 / 3,472 / 2,604

Baselines: yesterday (0a), same day last week (0b), 7-day average (0c), 14-day average (0d)
Metrics: MAE (main), RMSE, MAPE on $y > 0$, R^2

Performance Comparison

Table: Validation and test results (test after refitting on train+validation)

Method	Validation MAE	Test (1–12 August) MAE	RMSE	MAPE (%)	R^2
0a. Yesterday (`lag_1`)	2.666	2.740	3.986	70.9	0.469
0b. Same day last week (`lag_7`)	2.648	2.627	3.779	67.7	0.522
0c. 7-day average	2.182	2.168	3.114	54.8	0.676
0d. 14-day average	2.125	2.126	3.065	54.1	0.686
1a. Linear Regression (`lag_1` only)	2.563	2.642	3.724	69.4	0.536
1b. Linear Regression (38 columns)	1.999	2.078	2.991	53.9	0.701
2. Ridge ($\alpha = 0.1$)	1.999	2.078	2.991	53.9	0.701
3. KNN ($k = 50$, history + day of week)	1.985	2.043	2.988	50.8	0.702
4. Decision Tree (depth 7)	1.992	2.053	2.990	51.0	0.701
5. Random Forest (depth 8)	**1.963**	**2.023**	**2.938**	**50.5**	**0.711**

Key Results:

- **Random Forest**: test MAE = 2.023, $R^2 = 0.711$, **4.8%** lower MAE than the strongest baseline (14-day average, 2.126)
- The five ML models keep the same ranking on validation and test
- Yesterday $\rightarrow$ 14-day average already cuts validation MAE from 2.666 to 2.125; the best model adds less (to 1.963)

Hyperparameter Study I: KNN Feature Sets

Table: Validation MAE of the KNN regressor by feature set and number of neighbours k

Feature set	$k = 10$	$k = 30$	$k = 50$	$k = 100$	$k = 200$
All 38 columns	2.382	2.337	2.329	2.338	2.412
4 history columns	2.152	2.100	2.088	2.083	2.087
4 history columns + day of week	2.072	2.002	**1.985**	1.996	2.019

Key Findings:

- With all 38 columns KNN is worse than the 14-day average (2.329 vs 2.125): 27 binary attribute columns distort the distance
- History + day of week gives the best result: **1.985 with** $k = 50$, second-best model on validation
- Small k gives noisy averages, large k over-smooths; $k = 50$ lies inside the grid
- For distance-based methods, choosing the features matters as much as choosing the algorithm

Hyperparameter Study II: Tree Depth

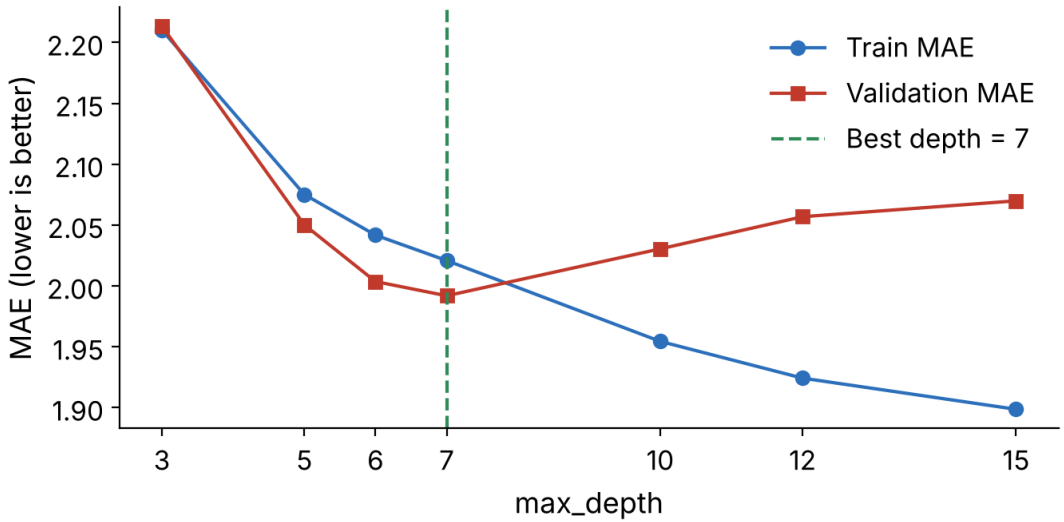


Figure: Decision Tree: train and validation MAE by `max_depth` (`min_samples_leaf` = 20)

Table: RF validation MAE

Depth	6	8	10	12
MAE	1.980	**1.963**	1.965	1.968

Key Findings:

- Train MAE keeps falling (2.210 → 1.899); validation MAE is lowest at depth 7 (1.992), then rises: overfitting
- Averaging 100 trees lowers the error further (1.963 at depth 8)

Hyperparameter Study III: Ridge and Multicollinearity

Table: Validation MAE of Ridge Regression by α

α	0.1	1	10	100	1000	10000
MAE	**1.999**	**1.999**	**1.999**	2.001	2.013	2.108

Table: Coefficients of the history features (standardised inputs)

Feature	Linear Regression	Ridge ($\alpha = 0.1$)	Ridge ($\alpha = 1000$)
`lag_1`	0.493	0.493	0.613
`lag_7`	0.207	0.207	0.410
`avg_7_days`	0.811	0.811	1.277
`avg_14_days`	2.696	2.696	1.799

Key Findings:

- `avg_7_days` and `avg_14_days` are highly correlated (0.983)
- A strong penalty spreads the weight across correlated features, at a small cost in MAE
- With $\approx$ 20,000 rows and 38 columns, least squares does not overfit $\Rightarrow$ no gain in accuracy

Reliability of the Improvement

Table: Daily test MAE

Date	RF	14-day	Win?
01/08	1.887	1.889	Yes
02/08	1.650	2.424	Yes
03/08	1.786	1.935	Yes
04/08	2.190	2.183	No
05/08	2.006	2.001	No
06/08	2.513	2.781	Yes
07/08	1.919	1.787	No
08/08	2.151	2.121	No
09/08	2.182	2.122	No
10/08	1.720	1.911	Yes
11/08	2.041	2.108	Yes
12/08	2.232	2.250	Yes

RF wins on 7 of 12 days; losses are small

Coin-flip test: if both methods were equally good, each day would be a fair coin

- 10,000 simulations of 12 flips: **39.1%** have ≥ 7 wins
- Passing the 5% level needs ≥ 10 wins:

$$\frac{\binom{12}{10}+\binom{12}{11}+\binom{12}{12}}{2^{12}} = \frac{79}{4096} \approx 1.9\%$$

Conclusion: the RF shows potential, but 12 days are not enough evidence to claim it is better

Error Analysis I: Actual vs Predicted

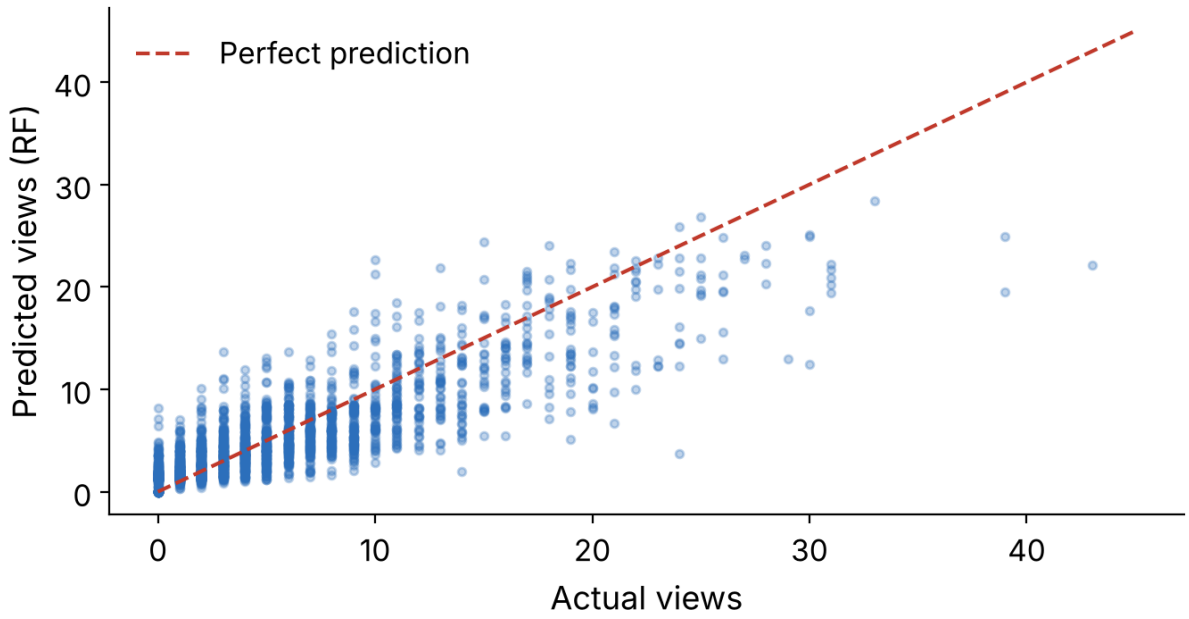


Figure: Actual vs predicted test views (RF); one point = one product on one day

Key Findings:

- For high actual values most points lie **below** the diagonal: the model pulls forecasts towards typical values
- Unusually popular days are under-predicted

Error Analysis II: Daily Totals and Categories

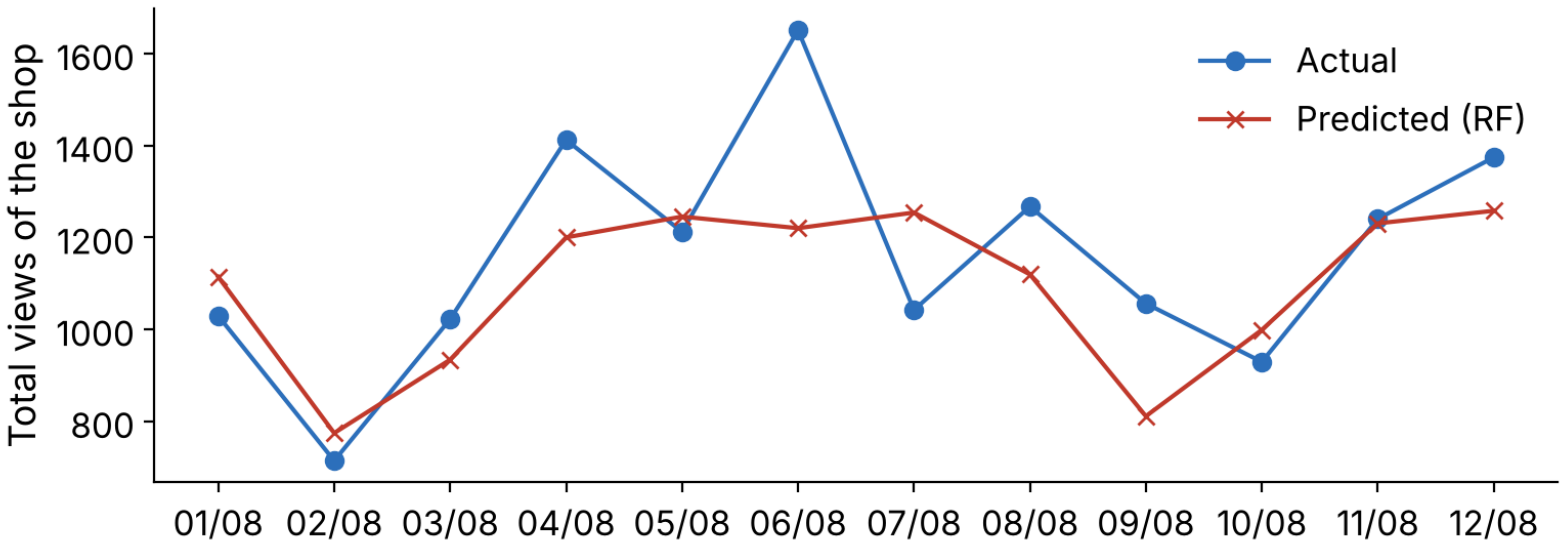


Figure: Total daily views of the shop on the test period

Table: Test error of the RF by category

Category	Mean views	MAE	Bias
Trousers	7.882	2.546	−0.460
Skirts	6.992	2.526	−0.261
Blouses	4.527	1.990	−0.352
Sale	3.971	1.570	−0.203

- Misses the 6 August spike (1,651 actual vs 1,220 predicted): no feature announces it
- Negative bias in all categories: slight under-forecast in August
- Without the 4 inactive products the gain is still 4.8%

Feature Importance

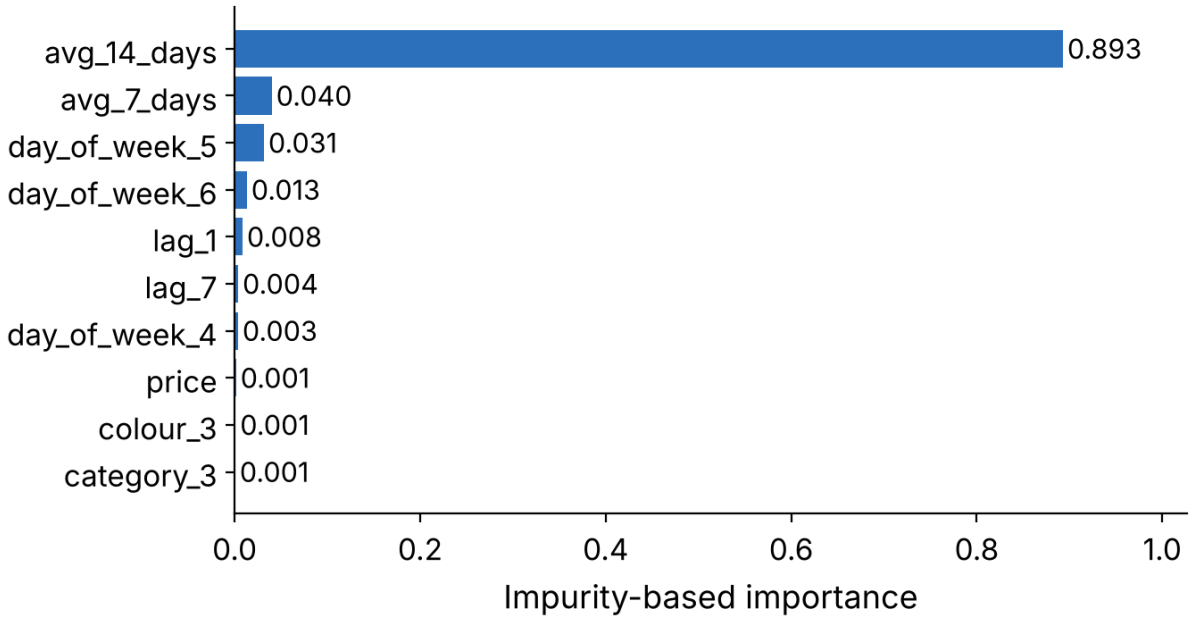


Figure: Top 10 features of the final Random Forest (impurity-based)

- `avg_14_days` dominates (0.893), then `avg_7_days` (0.040), Sat (0.031), Sun (0.013)
- Consistent with the strong 14-day baseline and the weekly cycle in EDA
- Caveats: correlated features share importance; it shows what the model *uses*, not what *causes* views [7]

Leakage Stress Test

Experiment: recompute `avg_7_days` **without** `shift(1)` and retrain the same Random Forest

- Correct: `x.shift(1).rolling(7).mean()`
- Leaky: `x.rolling(7, min_periods=1).mean()` (includes the forecast day)

Table: Effect of forgetting the one-day shift on the test scores

Random Forest	**MAE**	**RMSE**	**MAPE (%)**	R^2
Correct (with `shift(1)`)	2.023	2.938	50.5	0.711
Leaky (without `shift(1)`)	1.739	2.530	43.5	0.786

Key Findings:

- The leaky model looks much better (R^2 0.786), but the score is fake: nobody knows tomorrow's views tonight
- One missing line of code would overstate the result in the report and fail in deployment

Classification Results

Table: Classification view, sorted by validation F1 ($\tau = 7$ views)

Approach	Validation F1	Test Precision	Test Recall	Test F1
4. Decision Tree (depth 7)	**0.741**	0.782	**0.622**	0.693
R. RF regressor + threshold	0.734	0.792	0.615	0.693
3. KNN ($k = 50$)	0.715	0.788	0.602	0.683
5. Random Forest (depth 8)	0.715	0.796	0.619	**0.697**
1. Logistic Regression	0.711	**0.816**	0.579	0.678
0. Rule: 14-day average $\geq \tau$	0.695	0.760	0.618	0.682
2b. Gaussian NB (4 history columns)	0.695	0.784	0.573	0.662
2a. Gaussian NB (38 columns)	0.628	0.715	0.555	0.625

Key Findings:

- The selected Decision Tree reaches test F1 = 0.693, only ≈ 0.01 above the simple rule (0.682)
- Naive Bayes with 38 columns is weakest: correlated and one-hot columns break the independence assumption
- The RF regressor + threshold is as good $\Rightarrow$ regression already answers the yes/no question

Discussion: Comparison with the Common Approach

Table: The common `price2` task vs our forecasting task

Aspect	**Common `price2` task**	**Our forecasting task**
Target	Known price level	Tomorrow's views (unknown)
Split	Random clicks	By time (train / val / test)
Reference	None	Strongest naive baseline
Leakage	`price` gives the label	Checked by hand + stress test
Headline score	100% / 92.91%	MAE 2.023, R^2 0.711
On new data	44.1% (below 51.2%)	4.8% better than baseline
Business use	None	Daily merchandising

Key Messages:

- A modest gain measured honestly is worth more than a perfect score caused by leakage
- Most of the accuracy comes from **representing the history well**, not from model complexity
- Linear Regression (test MAE 2.078) is a reasonable simpler alternative if interpretability matters

Conclusions

Summary:

- We forecast tomorrow's views of 217 products from the *e-shop clothing 2008* clickstream
- The Random Forest, selected on validation, reaches test MAE 2.023 and R^2 0.711, **4.8%** better than the 14-day average
- It wins on 7 of 12 days (39.1% by chance): promising but not yet conclusive

Key Technical Contributions:

1. **Re-examination of** `price2`: three experiments show leakage and memorisation (44.1% on unseen products)
2. **Leak-free pipeline**: product–day panel with true zeros, past-only features checked by hand, time-based split, single test run
3. **Honest evaluation**: strongest baseline, day-by-day coin-flip test, error analysis, leakage stress test (R^2 0.711 vs fake 0.786), classification view

Limitations and Future Work

Limitations:

- Views are a proxy for demand; the data contain no purchases
- Only 4.5 months of 2008 data and a 12-day test period
- No information on promotions, holidays or campaigns; display position is fixed

Research Directions:

1. **Rolling-origin evaluation**: several test windows for more reliable conclusions [5]
2. **Richer features**: shop-level traffic, trend and dispersion features, session information
3. **Count-aware models**: Poisson regression and gradient boosting
4. **Business alignment**: 7-day horizon for stock; tuned classification threshold; sales if purchase data exist

Team Contribution

Table: Contribution of each member

Member	**Main tasks**	**Share**
Ho Huyen Trang	Parts 1–3: loading, cleaning, `price2` re-examination Report: Chapters 1–2	33,33%
Ta Van Sang	Parts 4–5: panel, EDA, features, leakage checks Report: Chapter 3	33,33%
Bui Quang Minh	Parts 6–9: models, tuning, evaluation, classification Report: Chapter 4, conclusion; slides	33,33%

Key References

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Thank You!